Title: Can Herding Explain Market Crashes? Reproducing the Lux–Marchesi Model With Today's S&P 500 Data.

Scientific Question: Using daily S&P 500 market returns with 1% movements, Black Monday when the market crashed 20%, and under a bell curve model that should not have happened in 10^86 years, show that these crashes happen within a few decades. The question is: if the news is normally distributed, does herding create fat tails and the volatility ranges seen in markets? Using Lux and Marchesi paper, it claimed it could be based on Kirman's model of ants herding in colonial behavior, and now I want to replicate it with today's S& P 500 data and compare it with the measurement of the real market that the model should match, and compare the herding with inverse cubic to show the fat tail.

What will I build: I will build an agent-based simulation web app that helps to simulate the stock market to test whether herding among traders explains more of why the market crashes than the normal bell curve predicts. I will first try to replicate Kirman's ant herding model to check against his math and see if I am on the right track. Then I would scale it to be on the Lux-Marchesi model of the 500 traders switching from following the true value of price to following based on trends and the return statistics, compared side by side with real S&P 500 data. I will use the news that will help to see how it affects normal bell curve distribution. In this web model, people are able to adjust the variables like the amount of herding, the strength of herding, and watch crashes and how they fluctuate in clusters.

Checking: I would check my model by comparing the extreme move values and volatility and seeing how it matches up to the S&P 500, and that no trend follower has a bell curve model. Where I will take beyond the paper is by trying to fix any weakness from that time that causes volatility to not be accurate. With zero chartists, my model will produce a normal bell curve distribution; this will be my base case to act on any errors and see the trader interaction that produces that fat tail. The fat tail that we should expect with a chartist is going to be around 2.6-3.

Data and Paper:

Lux, T. & Marchesi, M. (1999). Scaling and criticality in a stochastic multi-agent model of a financial market. *Nature* 397, 498–500. Supporting: Kirman (1993), *QJE*; Gopikrishnan et al. (1998), *Eur. Phys. J. B*.

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